SULAV REGMI

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PROFESSIONAL SUMMARY

Business Analytics and Data Science professional with 2+ years of experience in healthcare analytics, data engineering, machine learning, and AI-driven solutions. Proven expertise in analyzing large-scale EHR and claims datasets, developing predictive models, automating data workflows, and delivering actionable business insights. Selected as a top performer in the SAS Curiosity Cup 2026 and invited to present at SAS Innovate 2026. Skilled at translating complex data into scalable analytical solutions that support operational and strategic decision-making.

TECHNICAL SKILLS

- **Programming & Analytics**: Python (Pandas, NumPy, Scikit-Learn), SQL, R
- **Machine Learning**: Regression, Classification, Gradient Boosting, Neural Networks, Feature Engineering, Model Evaluation
- **AI & Agentic Systems**: Google Gemini, Vertex AI, Google CX Agent Studio, Multi-Agent Systems, Retrieval-Augmented Generation (RAG)
- **Data Engineering**: ETL Development, PySpark, MLflow, API Integration, Data Validation, Workflow Automation
- **Visualization & BI**: Power BI, Tableau, SAS Viya Workbench, Data Storytelling
- **Databases**: PostgreSQL, MySQL, MongoDB
- **Cloud & Tools**: Google Cloud Platform, Git, Hugging Face Spaces, Apify, n8n
- **Domain Expertise**: Healthcare Analytics, HEDIS Quality Measures, Regulatory Reporting, Quality Improvement Analytics

PROFESSIONAL EXPERIENCE

University of North Texas **Aug 2025 - May 2026**

Instructional Assistant *Denton, TX*

- Supported graduate-level Business Analytics coursework and helped manage course logistics for over 200 students, ensuring timely access to materials and smooth class operations.
- Mentored students in Python programming, predictive modeling, machine learning, and analytics project development, helping them successfully complete projects and strengthen their technical skills.
- Assisted faculty with grading, technical support, and evaluation of analytics assignments.

Cotiviti Nepal Pvt. Ltd. **Jul 2022 - Aug 2024**

Software Engineer / Data Analyst *Kathmandu, Nepal*

- Integrated and analyzed millions of healthcare records from EHR and claims systems using SQL, in-house tools and statistical methods, enabling payer and provider clients to meet HEDIS quality measurement deadlines and improve reporting compliance
- Developed Python and SQL data pipelines in Git-managed repositories for cleaning, validation, standardization, and transformation of large-scale healthcare datasets, cutting processing time by half and ensuring data readiness for downstream analytics
- Automated reporting and validation workflows, reducing manual effort by approximately 50% while improving delivery speed and data consistency.
- Produced recurring analytical reports and dashboard outputs supporting healthcare quality reporting, compliance monitoring, and internal decision-making.
- Collaborated with cross-functional teams to identify data quality issues and improve reporting accuracy across multiple client engagements.

Smarten Technologies Pvt. Ltd. **Apr 2022 - Jun 2022**

Software Engineering Intern *Kathmandu, Nepal*

- Assisted in developing enterprise applications on the Odoo platform by writing Python modules and configuring PostgreSQL databases, enabling faster deployment of new features
- Contributed to web development using HTML, CSS, and Odoo's web framework; performed system testing and documented results, which helped identify bugs and improve system reliability.
- Supported delivery of software enhancements that improved user experience and operational efficiency.

PROJECTS

Frontline Forecast: Hospital Length-of-Stay Prediction | Python | SAS Viya Workbench | LightGBM | Neural Networks

- Developed predictive models for hospital length-of-stay forecasting and bed allocation optimization using a dataset of more than 101,000 patient records across 130 hospitals.
- Built regression, classification, and uncertainty estimation models; achieved ROC-AUC of 0.837 and classification accuracy of 78.3%.
- Identified medication burden as the strongest predictor of prolonged hospital stays.
- Selected as a top-performing team in SAS Curiosity Cup 2026 and invited to present findings at SAS Innovate 2026.

Manufactured Home Price Prediction | Python | LightGBM | Streamlit | Hugging Face

- Built an end-to-end machine learning pipeline using 66,000+ housing records spanning 2014-2024.

- Achieved R^2 of 0.806 through feature engineering, anomaly detection, and model optimization.
- Deployed a production-ready Streamlit application on Hugging Face Spaces for real-time property valuation.

AI Tenant Maintenance Triage System | Vertex AI | Gemini | Google CX Agent Studio
- Designed and implemented a multi-agent AI platform for automated tenant maintenance request handling.
- Developed intelligent routing, vendor recommendation, escalation management, and human approval workflows.
- Improved evaluation quality scores from 79% to 93% through iterative prompt engineering and model optimization.

Paw Connect AI: AI Receptionist for Small Businesses | Dialogflow ES | Twilio | Google Apps Script
- Built a voice-enabled AI receptionist solution capable of handling bookings, FAQs, and customer inquiries.
- Automated appointment collection and staff notification workflows.
- Demonstrated over 90% cost reduction compared to traditional receptionist staffing models.

Rice Quality Analysis Using Machine Learning | Python | Flask | Scikit-Learn
- Developed machine learning classification models for automated rice quality assessment.
- Collaborated with a government-affiliated food management organization in Nepal.
- Deployed the best-performing model through a web application to improve classification accuracy and reduce post-harvest waste.

EDUCATION

University of North Texas **May 2026**
Master of Science, *Business Analytics*
- **Coursework:** AI for Business, Predictive Analytics, Data Mining, Data Visualization

Tribhuvan University **Jul 2022**
Bachelor of Engineering, *Computer Engineering*

ACHIEVEMENTS

- **Top Performer:**SAS Curiosity Cup 2026
- **Invited Presenter:**SAS Innovate 2026
- Presented healthcare analytics research focused on predictive hospital operations and resource optimization